Evaluating a Multi-Task Pipeline for Breast Ultrasound Segmentation and Classification

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***Abstract*—Breast ultrasound (BUS) is widely used for breast cancer diagnosis, but speckle noise and unclear lesion boundaries make automated analysis difficult. Existing deep learning pipelines typically handle segmentation and classification separately, requiring two encoders and two forward passes at inference. We built a multi-task learning (MTL) framework with a shared ResNet34 encoder that outputs a segmentation mask and a diagnostic label in a single forward pass. On the BUSI dataset (780 images, three classes), the MTL model reached 94.02% accuracy and a macro F1 of 93.22%, outperforming the dedicated classifier on both metrics. Segmentation results were mixed: IoU improved from 63.71% to 66.85%, while Dice fell from 77.78% to 73.96%, suggesting the joint model improves overall localization for most cases but exhibits higher variance on extreme outliers.**

***Index Terms*—Breast Ultrasound, Multi-task Learning, Tumor Segmentation, Image Classification, Deep Learning, Computational Efficiency.**

I. Introduction

Breast cancer is a common disease where early detection is critical for effective treatment. Breast ultrasound (BUS) is widely used because it is safe and affordable, but its images are often noisy with unclear boundaries, making diagnosis challenging. While deep learning methods have shown strong performance, most existing approaches focus on a single task, such as segmentation or classification, and require separate models.

Multi-task learning (MTL) allows a single model to handle both tasks at the same time, which can reduce computational cost and simplify deployment. However, sharing a model across tasks may also create conflicts that affect performance. Therefore, this study explores the effectiveness of an MTL framework for BUS analysis and examines the balance between performance and efficiency.

The main contributions are as follows:

- We design and train an MTL model with a shared ResNet34 encoder that segments tumors and predicts diagnostic class on the BUSI dataset.
- We use Optuna to automatically balance the segmentation and classification losses, removing the need for manual weight tuning.
- We compare the MTL model against dedicated single-task baselines on segmentation (Dice, IoU) and classification (accuracy, F1, precision, recall) metrics, and report the inference efficiency gain from using a shared backbone.

II. Related Work

A. Breast Ultrasound Classification

CNNs have been the dominant approach for BUS tumor classification. Das et al. [1] fine-tuned a ResNet backbone on BUS images and reported competitive accuracy on standard benchmarks, while Jabeen et al. [2] showed that EfficientNet transfers well to this domain with limited training data. A shared weakness of these models is that they output only a class label. The network has no explicit mechanism to focus on the lesion region, so classification decisions can be influenced by irrelevant background texture — a known issue in ultrasound where speckle patterns vary across machines and operators.

B. Breast Ultrasound Segmentation

U-Net and its variants remain the standard choice for lesion segmentation in BUS. The skip connections in U-Net preserve spatial detail that a plain encoder-decoder would lose, which matters when tumor margins are faint. Attention-gated extensions [3], [4] improve this further by suppressing activations in non-lesion regions, leading to cleaner boundary predictions. These models localize well, but they produce no malignancy estimate. In a real screening workflow, a radiologist would still need a separate classification step to act on the segmentation output.

C. Multi-Task Learning for Joint Analysis

Combining segmentation and classification under one model is a natural response to the above gap. Zhou et al. [5] were among the earlier groups to test this for BUS, and their main finding was that the segmentation branch functions as a soft attention signal: forcing the network to delineate the tumor boundary also encourages the classification head to concentrate on lesion features rather than background. Chowdary et al. [6] reached a similar conclusion using a shared encoder-decoder design. Both works, however, treated loss balancing as a fixed hyperparameter chosen by hand, and neither explicitly reported parameter counts or inference cost — aspects that matter when the target environment is a low-resource clinic.

Our work extends these findings in two ways: (1) automating loss weight selection via Optuna hyperparameter search, and (2) explicitly comparing the shared backbone's efficiency and performance trade-offs against single-task baselines.

III. Methodology

A. Network Architecture

The proposed system uses a shared-encoder design in which one backbone serves both the segmentation and classification branches. Fig. 1 shows the overall layout.

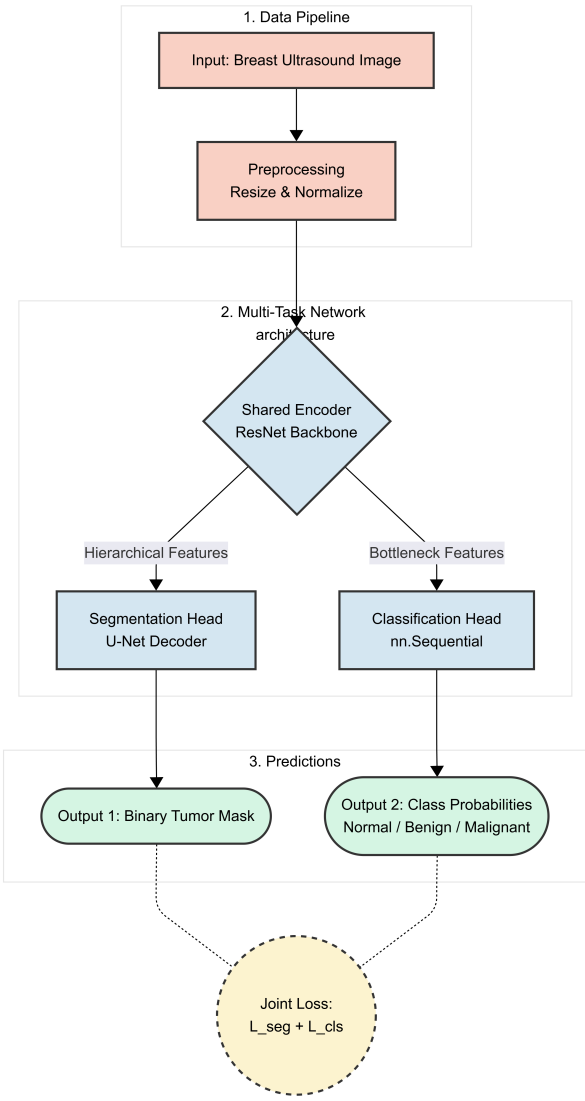

Fig. 1. Architecture of the proposed multi-task framework for breast ultrasound segmentation and classification.

Shared Backbone. A ResNet34 pre-trained on ImageNet serves as the encoder. It produces four feature maps at progressively reduced spatial resolutions, which are consumed by both downstream branches. Using a pre-trained backbone is important here because the BUSI dataset is relatively small (780 images); training a deep encoder from scratch on this scale tends to overfit.

Segmentation Head. The segmentation branch follows a U-Net-style decoder. Skip connections from each encoder stage are concatenated with the corresponding decoder output, preserving fine-grained spatial detail that would otherwise be lost during downsampling. A sigmoid activation on the final layer produces a pixel-wise probability map, which is thresholded at 0.5 to obtain the binary mask.

Classification Head. The classification branch operates on the deepest encoder feature map (8×8 spatial resolution for a 256×256 input). Global Average Pooling (GAP) collapses the spatial dimensions into a single feature vector, which is then passed through a fully connected layer with a softmax activation to predict probabilities over three classes: Normal, Benign, and Malignant. GAP was preferred over a flattening operation because it substantially reduces the parameter count and provides mild regularization.

B. Loss Functions

Each branch is supervised by its own loss term.

The segmentation loss combines Binary Cross-Entropy (BCE) and Dice Loss in equal parts:

$$\mathcal{L}_{seg} = 0.5 \cdot \mathcal{L}_{BCE} + 0.5 \cdot \mathcal{L}_{Dice} \tag{1}$$

BCE penalizes individual pixel errors, while Dice Loss directly optimizes the overlap ratio between the predicted mask and the ground truth. Their combination is common in medical segmentation because it handles the severe foreground-background imbalance that arises when tumors occupy a small fraction of the image.

For classification, Categorical Cross-Entropy is used:

$$\mathcal{L}_{cls} = -\sum_{c=1}^{3} y_c \log(\hat{y}_c) \tag{2}$$

where y_c is the one-hot ground-truth label and $\hat{y}_c$ is the predicted probability for class c.

C. Loss Balancing via Optuna

The two loss terms are combined into a single training objective:

$$\mathcal{L}_{total} = \alpha \cdot \mathcal{L}_{cls} + (1 - \alpha) \cdot \mathcal{L}_{seg} \tag{3}$$

The scalar $\alpha \in (0, 1)$ controls how much weight the training signal from each task carries. Setting α by hand is unreliable, a value that works on the validation split may not generalize, and grid search over this range is expensive. We therefore used Optuna to search for the value of α that maximizes the harmonic mean of the validation Dice score and the Macro F1-score:

$$HM = \frac{2 \cdot \text{Dice} \cdot \text{F1}}{\text{Dice} + \text{F1}} \tag{4}$$

The harmonic mean was chosen as the objective because it is sensitive to the weaker of the two scores: a model that excels at segmentation but performs poorly at classification will receive a low HM, which steers the search away from degenerate solutions that collapse to a single-task behavior.

IV. Experimental Setup

A. Dataset

All experiments use the publicly available Breast Ultrasound Images (BUSI) dataset [7], collected in 2018 from 600 female patients aged 25–75. The dataset contains 780 PNG images at an average resolution of 500×500 pixels, split across three clinical categories: 437 benign, 210 malignant, and 133 normal cases. Each tumor image comes with at least one ground-truth segmentation mask, bringing the total file count to 1578. The class distribution is summarized in Table I.

TABLE I
Image and mask distribution in the BUSI dataset.

Class	Images	Images + Masks
Benign	437	891
Malignant	210	421
Normal	133	266
Total	**780**	**1578**

B. Preprocessing and Augmentation

All images and masks were resized to 256×256 pixels and normalized with ImageNet mean and standard deviation, since the backbone was pre-trained on that distribution. We applied online augmentation during training using the random horizontal flipping, rotation up to ± 15, and random brightness and contrast adjustment. No augmentation was applied at validation or test time. Normal-class images were included in the classification branch but excluded from segmentation loss computation.

C. Data Splits

To keep the test set completely independent from all tuning decisions, we used a two-stage split. First, 15% of the full dataset was held out as a fixed test set. The remaining 85% was then divided 80/20 into training and validation sets. Stratified sampling was applied at both stages to preserve the original class ratios, given the imbalance between benign, malignant, and normal cases.

D. Implementation Details

The framework was implemented in PyTorch using the Segmentation Models PyTorch (SMP) library for the U-Net decoder. Training was run on an NVIDIA T4 GPU (16 GB VRAM) in Google Colab.

The Adam optimizer was used with an initial learning rate of 10^{-4}. A `ReduceLROnPlateau` scheduler halved the learning rate after 4 epochs without validation improvement. Batch size was set to 16 and training ran for 80 epochs across all experiments — the same configuration was used for both single-task baselines and the multi-task model to keep comparisons fair.

E. Loss Weight Search

Finding a good value for α in

$$\mathcal{L}_{total} = \alpha \cdot \mathcal{L}_{cls} + (1 - \alpha) \cdot \mathcal{L}_{seg} \tag{5}$$

is non-trivial because the two loss terms operate on different scales and the optimal balance is dataset-dependent. We ran 20 Optuna trials over $\alpha \in [0.1, 0.9]$, using the harmonic mean of the validation Dice score and Macro F1-score as the objective.

F. Baselines

After fixing α, the multi-task model was compared against two single-task baselines trained under comparable conditions:

- **STL-Seg:** U-Net with a ResNet34 encoder, trained with the same segmentation loss, supervised only on labeled tumor images.
- **STL-Cls:** ResNet34 with a GAP head, trained with Categorical Cross-Entropy on all three classes.

Both baselines use the same ResNet34 weights, preprocessing pipeline, optimizer settings, and train/test split as the multi-task model.

G. Evaluation Metrics

Segmentation quality is measured by the Dice Similarity Coefficient (DSC) and Intersection over Union (IoU). Classification performance is reported as overall accuracy, macro-averaged precision, recall, and F1-score. Macro averaging was chosen over weighted averaging because the three classes are clinically distinct and no single class should dominate the metric.

V. Results and Discussion

A. Loss Weight Search

Table II lists the top Optuna trials. The search converged on $\alpha = 0.20$, which yielded the highest harmonic mean of 0.8190. The result is notable: a low α means the training signal is dominated by the segmentation loss, suggesting that on BUSI, pixel-level supervision provides richer gradients for the shared encoder than the classification objective alone. All subsequent experiments use $\alpha = 0.20$.

TABLE II
Top Optuna trials ranked by harmonic mean of Dice and Macro F1-score.

α	Harmonic Mean
0.20	**0.8190**
0.40	0.8123
0.90	0.8042
0.60	0.7964
0.70	0.7867

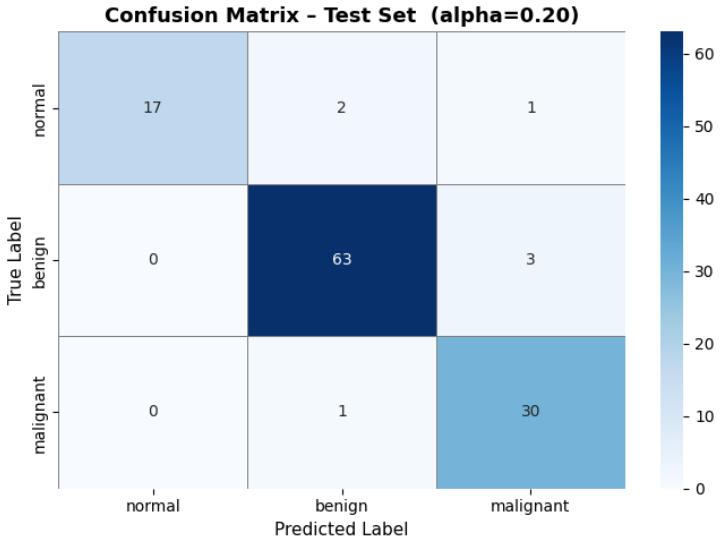

Fig. 2. Confusion matrix — MTL model.

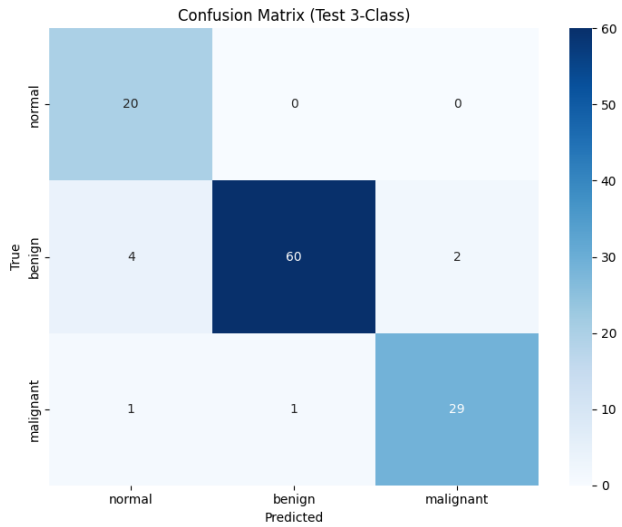

Fig. 3. Confusion matrix — STL-Cls model.

B. Classification

Table III compares classification performance on the held-out test set. The MTL model outperforms the dedicated classifier on accuracy (94.02% vs. 93.16%) and leads on precision (94.56% vs. 90.64%), while recall is slightly lower (92.41% vs. 94.82%). The recall gap is consistent with what task competition would predict: the shared encoder cannot allocate all of its capacity to class-discriminative features when it must also support boundary detection.

TABLE III
CLASSIFICATION PERFORMANCE ON THE TEST SET.

Model	Accuracy	Precision	Recall	F1
STL-Cls	93.16	90.64	**94.82**	92.31
MTL	**94.02**	**94.56**	92.41	**93.22**

All values in %. Precision, Recall, F1 are macro-averaged.

One plausible explanation is that the segmentation branch forces the encoder to attend to the lesion region, suppressing activations from background speckle that might otherwise push borderline cases toward a false positive prediction.

The confusion matrix in Fig. 2 shows the MTL model correctly classified 17 of 20 normal cases, 63 of 66 benign cases, and 30 of 31 malignant cases. Notably, no malignant case was misclassified as normal - an error that would delay treatment in a real workflow. The STL-Cls model made this error at least once, as seen in Fig. 3. Whether this difference holds on a larger test set is an open question, but the pattern is consistent with the precision improvement in Table III.

C. Segmentation

Table IV shows a mixed picture. The MTL model reached an IoU of 66.85%, a 3.14 point gain over STL-Seg (63.71%), but its Dice score (73.96%) fell short of the baseline (77.78%).

TABLE IV
SEGMENTATION PERFORMANCE ON THE TEST SET.

Model	Dice (%)	IoU (%)
STL-Seg	**77.78**	63.71
MTL	73.96	**66.85**

Dice is more sensitive to boundary precision than IoU. Because both metrics are macro-averaged, the simultaneous increase in IoU and decrease in Dice points to higher prediction variance across the test set: the MTL model produces better masks for most cases, but struggles on a small subset of ambiguous examples where boundary precision is heavily penalized. This suggests global context aids overall localization but does not consistently sharpen contour prediction.

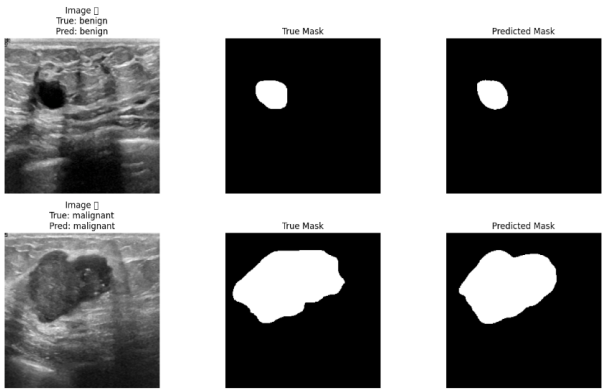

Fig. 4. Qualitative segmentation results.

Fig. 4 reflects this: the MTL masks cover the gross lesion extent well, but boundaries in irregular cases are occasionally less tight than the ground-truth annotation.

D. Computational Efficiency

Table V collects all metrics for direct comparison.

TABLE V
SUMMARY OF TEST-SET PERFORMANCE ACROSS ALL MODELS.

Model	Acc.	F1	Dice	IoU
STL-Cls	93.16	92.31	–	–
STL-Seg	–	–	**77.78**	63.71
MTL (proposed)	**94.02**	**93.22**	73.96	**66.85**

All values in %.

Running both baselines in deployment would require two separate forward passes and two sets of weights, one ResNet34 classifier and one ResNet34-backed U-Net. The shared backbone in the MTL model eliminates the duplicate encoder, so a single forward pass produces both the segmentation mask and the diagnostic label. This reduces memory footprint and inference cost compared to running the two baselines sequentially — a difference that can determine whether real-time operation is feasible on edge devices or workstations with limited VRAM.

VI. CONCLUSION

This paper asked whether a single shared-encoder network can replace two dedicated models for breast ultrasound analysis without sacrificing diagnostic reliability. On the BUSI dataset, the answer is mostly yes, though not uniformly across metrics.

On classification, the MTL model outperformed the standalone ResNet34 on accuracy (94.02% vs. 93.16%), precision (94.56% vs. 90.64%), and F1 (93.22% vs. 92.31%), with only a small recall gap (92.41% vs. 94.82%). On segmentation, the picture is mixed: IoU improved from 63.71% to 66.85%, but Dice fell from 77.78% to 73.96%. This divergence suggests that while the MTL model segments most images better, it struggles with higher variance on a few difficult outliers. The shared backbone also reduces inference to a single forward pass, lowering computational overhead compared to running two models sequentially.

The test set contains only 117 images, so per-class differences in the confusion matrices should not be over-interpreted. Future work should evaluate the framework on external BUS datasets to test generalization, and explore learned loss weighting as a replacement for the fixed α found by Optuna — particularly to see whether closing the Dice gap is possible without sacrificing the IoU and classification gains.

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